

A03 VENTILATION DESIGN | P

Intent : Minimize indoor air quality issues through the provision of adequate ventilation.

Summary : This WELL feature requires projects to bring in fresh air from the outside through mechanical and/or natural means in order to dilute human- and product-generated air pollutants.

Issue : Poorly ventilated spaces contribute to symptoms, such as headache, fatigue, dizziness, nausea, cough, sneezing, shortness of breath and eye, nose, throat and skin irritation. This is collectively called sick building syndrome (SBS).^{1,2} Poor ventilation is also linked to increased rates of employee absences, higher operational costs for businesses and decreased productivity in students.^{3,4} One U.S.-based study reported that the sick leave attributable to insufficient provision of fresh air in buildings is estimated to be 35% of total absenteeism.⁵ Therefore, the economic costs of SBS in under-ventilated buildings are significant and far exceed the energy-related cost savings.⁶⁻⁸

Solutions : Many indoor and outdoor sources of air pollution emit particulate matter and volatile organic compounds (VOCs) that can cause discomfort and trigger asthma and eye, nose and throat irritation. In order to maintain healthy indoor environments and acceptable air quality for building users, it is necessary to provide sufficient ventilation.^{9,10} In addition to proper HVAC system design, mechanically ventilated projects need to perform regular system maintenance, since inadequate maintenance is associated with reduced ventilation performance and a deterioration in indoor air quality and thermal conditions.¹¹ For naturally ventilated spaces, it is necessary to ensure that adequate outdoor air quality and noise levels are met.

PART 1 ENSURE ADEQUATE VENTILATION

For All Spaces:

Option 1: Mechanically ventilated spaces

For mechanically ventilated buildings, one of the following requirements is met:

- a. Newly installed ventilation systems are designed to meet the supply and exhaust rates set in one or more of the following ventilation guidelines, which must describe ventilation rates for at least 90% of the project area. The ventilation system is scheduled to be tested and balanced after project occupancy:
 1. ASHRAE 62.1-2010 or any more recent versions (Ventilation Rate Procedure or IAQ Procedure).¹²
 2. ASHRAE 62.2-2016.¹³
 3. EN 16798-1.¹⁴
 4. AS 1668.2-2012 or any more recent version.¹⁵
 5. CIBSE Guide A: Environmental Design, version 2007 or any more recent version.¹⁶
- b. Existing ventilation systems have been tested and balanced to meet supply and exhaust rates set in one or more ventilation guidelines listed above within the last five years.

OR

Option 2: Naturally ventilated spaces

For naturally ventilated buildings with no mechanical ventilation, the following requirements are met:

- a. One or more of the following design criteria, which must describe ventilation rates for at least 90% of the project area:
 1. Natural Ventilation Procedure in ASHRAE 62.1-2010 or any more recent version.¹²
 2. CIBSE AM10: Natural Ventilation in Non-Domestic Buildings (2005 or any more recent version) Section 2.4 – Natural ventilation strategies and Chapter 4 – Design Calculations.¹⁷
 3. AS 1668.4-2012 or any more recent version.¹⁵
 4. Any reference in Option 1, which describes natural ventilation procedures.
- b. Vents and windows used to meet the ventilation requirements in one of the standards mentioned above are permanently open or have controls to prevent their closure during periods of occupancy. (Operable windows not used in ventilation calculations may be user operated.)
- c. Outdoor air meets the following thresholds as an average for the previous year:
 1. PM_{2.5} less than 15 µg/m³.¹⁸
 2. PM₁₀ less than 30 µg/m³.¹⁸

OR

Option 3: Naturally ventilated spaces in areas with elevated particulate matter

Note: Projects pursuing this strategy are limited in WELL Certification level to Silver regardless of total points achieved.

For naturally ventilated buildings with no mechanical ventilation, the following requirements are met:

- a. One or more of the following design criteria, which must describe ventilation rates for at least 90% of the project area:

1. Natural Ventilation Procedure in ASHRAE 62.1-2010 or any more recent version.¹²
 2. CIBSE AM10: Natural Ventilation in Non-Domestic Buildings (2005 or any more recent version) Section 2.4 – Natural ventilation strategies and Chapter 4 – Design Calculations.¹⁷
 3. AS 1668.4-2012 or any more recent version.¹⁵
 4. Any reference in Option 1, which describes natural ventilation procedures.
- b. Vents and windows used to meet the ventilation requirements in one of the standards mentioned above are permanently open or have controls to prevent their closure during periods of occupancy. (Operable windows not used in ventilation calculations may be user operated.)
- c. Outdoor air meets the following thresholds as an average for the previous year:
1. PM_{2.5} less than 35 µg/m³.¹⁸
 2. PM₁₀ less than 70 µg/m³.¹⁸

OR

Option 4: Ventilation monitoring

One of the following carbon dioxide thresholds is met in occupiable spaces:

- a. 900 ppm or less.
- b. Not more than 500 ppm above outdoor levels.

Note : Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

Note :

Mixed mode projects must meet the requirements set for naturally ventilated spaces when mechanical system is not in use and the requirements for mechanical ventilation when the mechanical system is in use. The project must use mechanical ventilation when outdoor PM_{2.5} levels are above 15 µg/m³, based on a measuring device within 4 km of the project.

REFERENCES

1. Wargocki P, Wyon DP, Sundell J, Clausen G, Fanger PO. The effects of outdoor air supply rate in an office on perceived air quality, sick building syndrome (SBS) symptoms and productivity. *Indoor Air*. 2000;10(4):222-236. <https://onlinelibrary.wiley.com/doi/epdf/10.1034/j.1600-0668.2000.010004222.x>.
2. Daisey JM, Angell WJ, Apte MG. Indoor air quality, ventilation and health symptoms in schools: An analysis of existing information. *Indoor Air*. 2003;13(1):53-64. doi:10.1034/j.1600-0668.2003.00153.x
3. Haverinen-Shaughnessy U, Moschandreas DJ, Shaughnessy RJ. Association between substandard classroom ventilation rates and students' academic achievement. *Indoor Air*. 2011;21(2):121-131. doi:10.1111/j.1600-0668.2010.00686.x
4. Chan WR, Parthasarathy S, Fisk WJ, Mckone TE. Estimated effect of ventilation and filtration on chronic health risks in U.S. offices, schools, and retail stores. *Indoor Air*. 2016;26(2):331-343. doi:10.1111/ina.12189
5. Milton DK, Glencross PM, Walters MD. Risk of sick leave associated with outdoor air supply rate, humidification, and occupant complaints. *Indoor Air*. 2000;10(4):212-221. https://buildequinox.com/files/iaq/milton_vent_sick_rates.pdf.
6. Redlich CA, Sparer J, Cullen MR. Sick-building syndrome. *Lancet (London, England)*. 1997;349(9057):1013-1016. <https://www.ncbi.nlm.nih.gov/pubmed/9100639>.
7. Fisk WJ. Estimates of improved productivity and health from better indoor environments. *Indoor Air*. 1997;7(3):158-172. doi:10.1111/j.1600-0668.1997.t01-1-00002.x
8. Fisk WJ. How IEQ affects health, productivity. *ASHRAE*. 2002;44(5).
9. Sundell J, Levin H, Nazaroff WW, et al. Ventilation rates and health: Multidisciplinary review of the scientific literature. *Indoor Air*. 2011;21(3):191-204. doi:10.1111/j.1600-0668.2010.00703.x
10. Carrer P, Wargocki P, Fanetti A, et al. What does the scientific literature tell us about the ventilation-health relationship in public and residential buildings? *Build Environ*. 2015;94(P1):273-286. doi:10.1016/j.buildenv.2015.08.011
11. Hanssen SO. HVAC - The importance of clean intake section and dry air filter in cold climate. *Indoor Air, Suppl*. 2004;14(SUPPL. 7):195-201. doi:10.1111/j.1600-0668.2004.00288.x
12. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 62.1-2010: Ventilation for Acceptable Indoor Air Quality.; 2010.

13. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 62.2-2016: Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings. <https://www.ashrae.org/technical-resources/standards-and-guidelines/standards-addenda/addenda-to-standard-62-2-2016>. Published 2016.
14. European Committee for Standardization. EN 16798-1:2019 - Ventilation for Buildings. 2019. https://standards.cen.eu/dyn/www/f?p=204:110:0:::FSP_PROJECT,FSP_ORG_ID:41425,6138&cs=11EDD0CE838BCEF1A1EFA39A24B6C9890.
15. Standards Australia. AS 1668-2012 The Use of Ventilation and Air-Conditioning in Buildings. Sydney; 2012.
16. Chartered Institution of Building Services Engineers. CIBSE Guide A: Environmental Design. Chartered Institution of Building Services Engineers; 2007. doi:10.1016/B978-0-240-81224-3.00016-9
17. Chartered Institution of Building Services Engineers. AM 10 Natural Ventilation in Non Domestic Buildings.; 2005. <https://www.cibse.org/knowledge/knowledge-items/detail?id=a0q20000008I7m2AAC>.
18. World Health Organization. Air Quality Guidelines for Particulate Matter, Ozone, Nitrogen Dioxide and Sulfur Dioxide. Geneva, Switzerland; 2005. http://apps.who.int/iris/bitstream/handle/10665/69477/WHO_SDE_PHE_OEH_06.02_eng.pdf;jsessionid=CD44465Fsequence=1.